

Understanding Through Virtual Reality the Cinematic Expressionistic Architecture

The case study of The Cabinet of Dr Caligari

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ABSTRACT

This interdisciplinary research project explores the symbiotic relationship between architecture and cinema through the application of Virtual Reality (VR) technology. Focusing on the iconic film *The Cabinet of Dr Caligari* (1920) by Robert Wiene, renowned for its pioneering narrative and architecturally innovative film sets, the study employs VR to digitally reconstruct these lost film sets. By leveraging archival drawings and meticulous data acquisition, the project's iterative design process models the sets to near-physical accuracy. The resulting VR simulation offers a unique opportunity for scholars, architects, and film enthusiasts to immerse themselves in the expressionist era's spatial and atmospheric environments. Additionally, it advocates for the use of VR as a tool for the preservation and restoration of cultural heritage within the realm of film and architecture. The case study's significance lies in its contribution to the preservation of film culture and the architectural legacy of Expressionism.

CCS CONCEPTS

• **Human-centered computing** → Visualization; Visualization design and evaluation methods; • **Applied computing** → Arts and humanities; Architecture (buildings); Computer-aided design.

KEYWORDS

Architecture, Expressionism, Virtual Reality, Digital Cultural Heritage, Post-Phenomenology

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1 INTRODUCTION

Computer Aided Architectural Design (CAAD) and visualization methods are extensively employed in the domains of established arts, such as architecture, and the game design industry, to effectively communicate proposed buildings and structures. The

utilization of digital visualization extends beyond these fields, encompassing architecture, cinema, music, and the gaming realm, thereby enabling the audience to perceive artistic creations in a contemporary manner. Furthermore, virtual representations are not limited to visualizing modern and futuristic settings but are also leveraged to depict interpretations of historical environments. This includes portraying ruined or time-worn buildings, as well as geographical locations, which no longer exist in the present, and thereby immersing the viewer in a virtual world of the past. Architectural visualization utilizes a variety of techniques including two-dimensional (2D) representation of buildings, the enhancement of drawings, photography with collage or motion images, and three-dimensional (3D) techniques in which designers create physical, or since 1976 in the film *Futureworld* directed by Richard T. Heffron, virtual models in Computer-Generated Imagery (CGI) environments [1]. One of the most immersive technologies, if not the most, is Virtual Reality (VR).

This case study is an interdisciplinary research project, which examines the affinities between architecture and cinema through the use of VR, involving the visualization and reconstruction of some of the most significant expressionist stage sets in the history of architecture and cinematography. Older, classic films have an important role in cinematic history not only for their pioneering narratives but also for their film sets, some of which are recognized as architecturally innovative: *The Cabinet of Dr Caligari* (1920), *Metropolis* by Fritz Lang (1927) and a later generation of films including *Rear Window* (1954) by Alfred Hitchcock. This paper presents the design process of reconstruction in a VR simulation of the film sets that have been damaged beyond repair, and no longer physically exist, through the case study of *The Cabinet of Dr Caligari*.

Expressionism movement in architecture has a limited presence as physical structures and constructions. Expressionist architects tended to create their ideas within other media, such as painting, writing and the construction of film and theatre sets. As the existence of expressionist 3D elements is scarce, this research argues that expressionist stage sets are important for restoration purposes in an effort to preserve cultural heritage from a filmmaking perspective and an architectural standpoint. An important outcome of this study is the record of the methodology used for this reconstruction, as well as these recreated VR film sets that have been made available for public viewing, preventing these classic architectural and expressionist film designs from being lost or forgotten. Architects, theorists, film critics, scholars, and educational establishments at all levels of study will be able to access the VR archive of *The Cabinet of Dr Caligari*, and explore but also experience the film sets as near as possible as they were intended to be 'physically'. Contributing

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to preserving film culture via the opportunity of re-visiting these classic film sets and making them available for further analysis through the lens of an architecture-cinematography relationship, is a key benefit of this case study.

2 RELATED WORK

3D technology has proved to be a valuable tool for the reconstruction and preservation of cultural heritage multiple times in the past, establishing the term *Digital/Virtual Heritage* [2, 3]. This section includes only selected previous works, which have used similar techniques in their projects, such as 3D digital reconstruction based on the archival materials and presentation of VR simulations demonstrating 3D visualizations of cultural heritage artefacts. For example, M.G. Bevilacqua et al. used the available archival materials in photogrammetry for the 3D digital reconstruction of Palazzo di Cosimo de' Medici, in the Fortezza Vecchia site [4]. Another example is Basilica di San Giovanni in Conca in Milan, which was reconstructed based on the high-resolution digital survey of its ruins, providing a virtual reconstruction from different historical evidence [5]. Recently, VR has been introduced in digital heritage reconstruction in multiple ways. Either through an interactive visualization giving the user the ability to interact with the virtual environment [6], or through pre-rendered 360-degree videos such as Argyriou's et al. project about reserved artefacts in the city of Rethymno in Greece [7]. In addition, VR offers the user the incredible ability to explore the virtual space through real-time VR. Related projects, which operate real-time VR for virtual heritage reconstruction purposes, include St Augustine Church in La Laguna [8], St Augustine's Abbey in Canterbury [9] and the Sanctuary of Delphi in Greece [10]. The two last cited projects faced constraints due to the incomplete archival materials. Therefore, necessary interpretations and assumptions of the available data and the existing ruins had to be applied to preserve digitally cultural heritage [5, 6]. Similarly, this paper presents a VR technique which was applied to develop a digital reconstruction of Dr Caligari's film sets, a film which also presented limitations on its documentation with the difference that offers a real-time rendering VR simulation and the ability to walk through the virtual space.

3 ITERATIVE DESIGN PROCESS

The general aim of this study is to produce a VR simulation of Dr Caligari's film setting as faithful as possible to the original film sets and evaluate to what extent VR contributes to the understanding of its abstract spatial environment. Through the use of VR, the film sets are available for virtual observation in real-time, and the following objectives are essential to reach the general aim:

OB1. Use of the available archival data to obtain the necessary information for the virtual reconstruction of Dr Caligari's film setting (i.e. plans, elevations and sections of the film sets, measurements of the backdrops, painting patterns, etc.).

OB2. Proposition of methodological processes for the 3D digital reconstruction of the film sets and studio, which allows for elaborating and evaluating hypotheses.

OB3. Digitalization of wall paintings and cloning of the patterns to be used as a texture for the 3D modelling. This presents a necessary step for reconstructing the setting, aiding the photorealism,

presenting the film set closer to how it was and presenting the expressionistic atmosphere that existed on the stage.

OB4. Production of analytical representations, photorealistic visualizations, and documentation are other multifaceted applications used to analyze the 3D digital model further.

OB5. Creation of a VR simulation that can be viewed by (an HMD) VR headset. This allows the author and potential users to explore by walking into the studio's virtual reconstruction, including five of Dr Caligari's film sets. This is the most critical step for the dissemination of this project.

Considering these objectives, the hypothesis suggests that VR could be a useful tool for the digital recovery of obscure architecture, offering a better understanding of the expressionist film sets of Dr Caligari.

3.1 Integration with archival drawings and data acquisition

The virtual reconstruction of the film sets is based on detailed data compiled from historical materials found in the archives. The overall collection of archival materials is sourced from Deutsche Kinemathek and La Cinémathèque Française. The *Grafik Archive* in Deutsche Kinemathek has been the primary source of data, including a collection of 173 graphic documents in total. Of these, 125 are hand drawn by Hermann Warm, the architect of the film sets, and from those, 48 are architectural drawings, 21 of which contain technical architectural details of the film sets, including plans, elevations and sections measured on 1:50 scale. Archivists at Deutsche Kinemathek acknowledge incomplete documentation, particularly regarding specific production dates of those drawings. The Graphics archive's drawings, reconstructed by the architect of the film sets in the 1960s, are believed to greatly reflect those from the 1920s. Even if there are substantial grounds for believing that similar drawings existed in the 1920s, still this uncertainty is a limitation that cannot be neglected.

After establishing the authenticity and validity of the drawing collection by a team of specialist archivists, and inspired by Niek Turner's research methodology [11], the drawings were examined individually along with a *scene-by-scene* film analysis (Step 1a and Step 1b, Figure 1); a methodology that examines each drawing against how they relate to the film sets by inspecting each shot in the film and grouping the drawings into scenes (Step 2, Figure 1). The individual source analysis takes a qualitative approach, that is, a process of describing, evaluating, classifying, and commenting critically on each drawing of the archival collection. This process shows how Warm's drawings reveal the way the film sets were created and formed in each scene in the *Lixie-Film-Atelier-Weißensee GmbH* studio. However, a comprehensive analysis is limited by the lack of information in some drawings such as production dates, the props' design and dimensions. A variety of software was used, such as Adobe Premiere Pro, Adobe Photoshop, and Adobe Illustrator in the creation of the analytical diagrams. The building of the VR reconstruction requires film sets to be described by a complete set of drawings. Such film sets were selected after a thorough examination of the drawings (*individual source analysis*, in Step 3, Figure 1). Further scrutiny revealed which architectural drawings

Content Analysis - Diagram

Step 1a_Automatically detect shot changes in footage Method: Adobe Premiere Pro

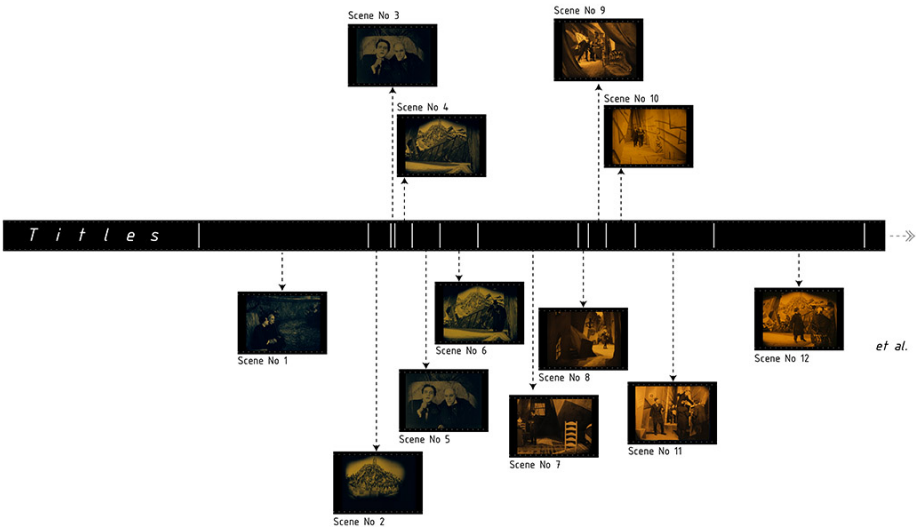


Footage from the film *The Cabinet of Dr. Caligari* (1920)

Adobe Premiere Pro's function called *Scene Edit Detection* automatically detects the original edit points or shots.



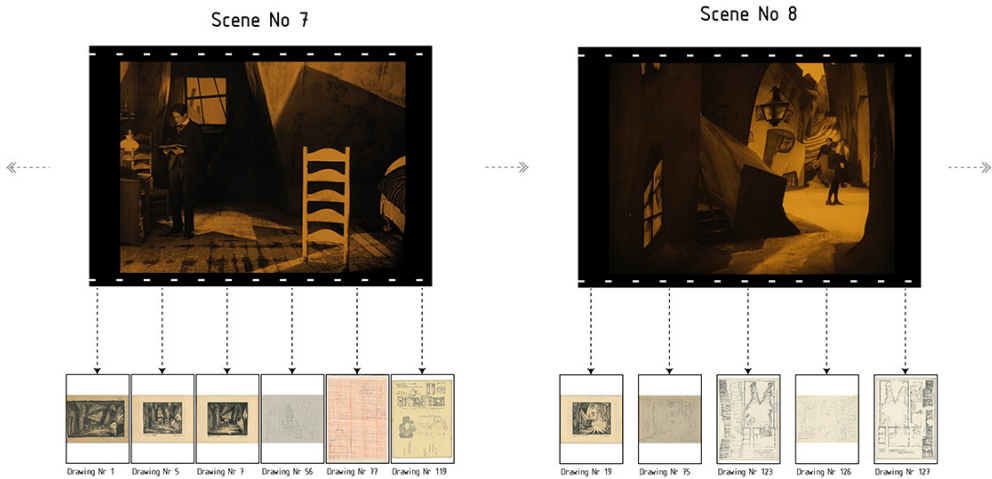
Step 1b_Separate the shots into scenes Method: Manually & Adobe Premiere Pro



I was able to group the shots with the same location and time into scenes by detecting the shots with the use of Adobe Premiere Pro. Therefore, I was able to define with accuracy when the scene changes. A scene is a unit of a story that develops at a specific location and time. If either of these changes, then a new scene is created. Adobe Premiere Pro was used in this stage to export the still frames from each scene.



Step 2_Link each scene with the related archival drawings Method: Archival Catalogue from Deutsche Kinematik & Manually



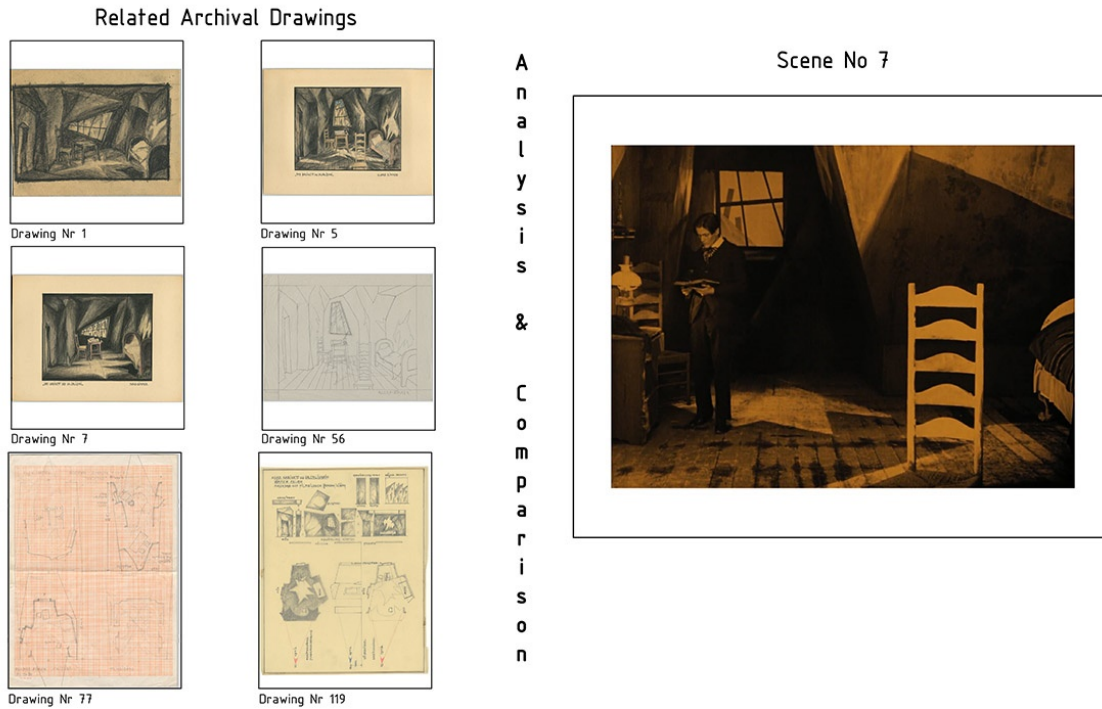
This systematic analysis follows the film's narrative and relates the scenes with the relevant drawings.





Step 3. Description and Analysis of the drawings used in each scene

Method: Adobe Premiere Pro, Illustrator, Photoshop



This methodology demonstrates A), how Hermann Warm's drawings were used for creating these scenes and for constructing film sets, and B), how the drawings were used in shooting the scenes sequentially.

Figure 1: Diagram methodology – Content analysis. Altered by the author. Deutsche Kinemathek – Hermann Warm Archiv.

and instructions were faithful to the film set, and which differed from their actual construction.

The study of the film sets started with examining their 1:50 scale plans, elevations, and sections. Converting and redrawing these architectural drawings into Computer Assisted Design (CAD) - compatible vector format allowed for their coherent understanding. This was achieved by using Autodesk®AutoCAD software. This was not only an efficient way to critically read and record each construction detail included in those drawings, but also a way to detect missing clues, such as the backdrops' supporting system, the backdrops' depth, the spotlights' position, and the furniture's detailed design, including their dimensions.

A plan (Figure 2) informs the author about the accurate positions of the backdrops and the furniture in the set, as well as the backdrops' width. Additional information regarding the backdrops' dimensions could be collected from the elevation, in which both length and width are available. By reading the annotations in the

elevation, the positions of the respective backdrops in the plan were determined as well as the instructions for the assembling process. Important information included in the section is the camera's position, which is a clue that was used as a reference to rescale the scanning drawing in AutoCAD into a 1:50 scale for printing.

These analyses also serve to demonstrate how the methodology could adapt to other film sets with different characteristics, but similar archival materials. Computer drawings (CAD) and software such as Photoshop, 3Ds Max, V-Ray and Unreal Engine 4.27 were used systematically for this 3D virtual reconstruction and visualization process. These tools proved to be valuable in graphic recording and analysis of these archival materials, adding to the better comprehension and presentation of the environment, and subsequently helping to evaluate the virtual reconstruction hypotheses.

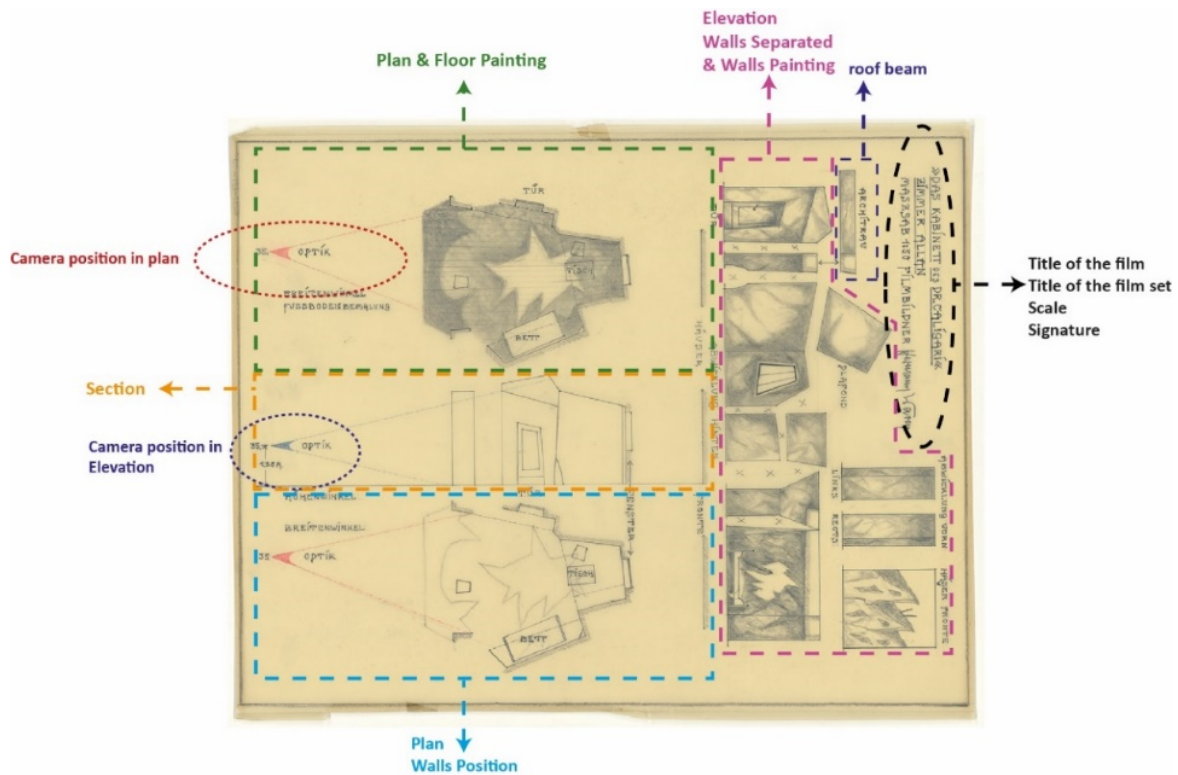


Figure 2: Data in the architectural drawings. Altered by the author. Deutsche Kinemathek – Hermann Warm Archiv.

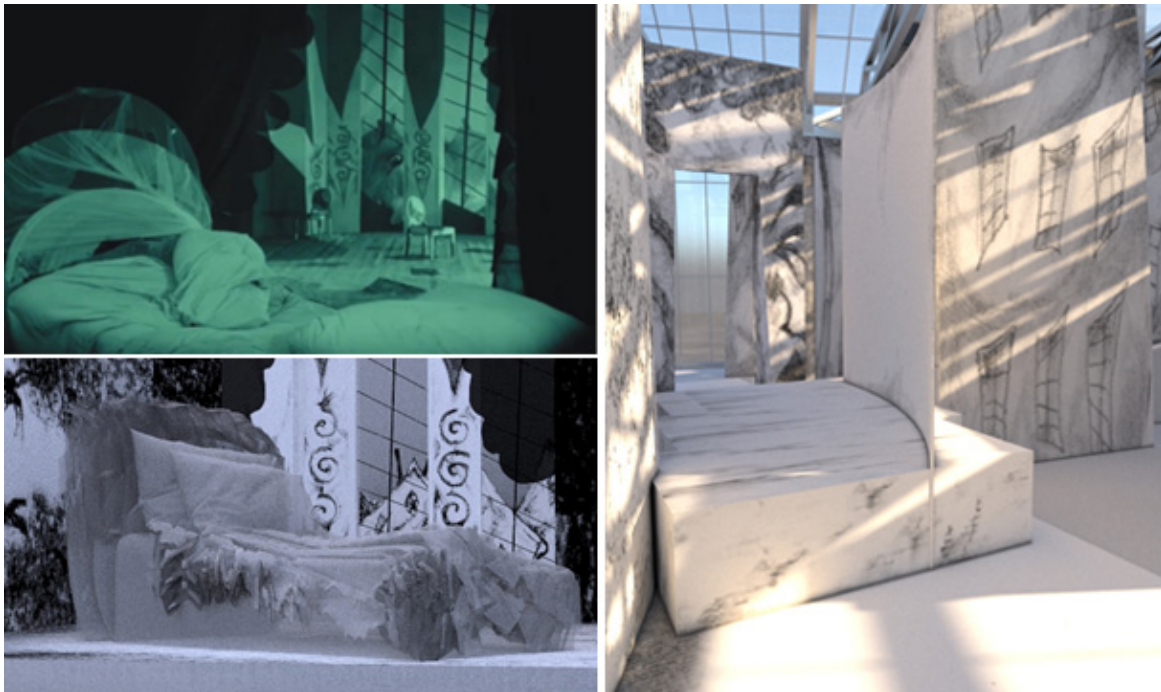


Figure 3: (a) Screenshot of Dr Caligari's film by Robert Wiene, 1920, (b) Render of the furniture with low opacity. Created by author using 3Ds Max and VRay, (c) View of the same set from the VR space. Created by Author.

4 (POST-) PHENOMENOLOGICAL ANALYSIS

The focal point to understand a space is the body inside it and how the individual feels and experiences this space is through self-movement. Taking this into consideration, the phenomenological approach has been proven as the most suitable methodology to be applied for evaluating this VR simulation based on my research. According to Husserl, to identify what we are perceiving now in an experience and to identify what there is a possibility to experience in the future has to be clarified using the technical terms *inner horizon* and *outer horizon* [12]. According to him, the *inner horizon* identifies our current observation of one side of things and the *outer horizon* refers to the possible future insights of other unseen sides. Hence, this thesis's results and conclusions consist of my *inner horizon* experience of Dr Caligari's expressionist architecture through watching the film and analyzing the drawings, which is the one side of observation currently available. Additionally, they also contain my *outer horizon* experience, which presents the unseen aspects such as the studio, and how the sets were placed into it, adding further insights into this expressionist construction as this time, my exploration is by participating in a 3D reconstruction of the film sets.

Patricia Locke, co-editor of Merleau-Ponty: Space, Place, Architecture, draws an analogy between VR and painting, seeing the latter 'as a portal to reality, a form of a virtual reality interface' [13]. Warm's drawings are media that display and describe Dr Caligari's film sets to the viewers. The film can visually transfer them to this expressionist atmosphere. However, both have the limitation of a 2D image, moving or not, which restricts the extended comprehension of the expressionist architecture. This VR simulation does actually place the viewers into the world of Dr Caligari, immersing them in the expressionist environment and letting them explore the studio with the film sets as has been described in a 2D form in the archival documents. This project enables the kinesthetic consciousness by adding the ability to move between the film sets irrespective of our actual bodily location. This is another one of the reasons that the phenomenology of Husserl and his theory of kinaesthesia can be applied and recognize the opportunities that this practical element offers to the author and other scholars. The expressionist architecture in *The Cabinet of Dr Caligari* has been considered as a paradigmatic and prototype of Expressionism. Architecture is one of the most spatial arts and it is important to secure humans' spatial existence. Through the use of VR, the spatial existence into the space of the film has been secured, giving the users the choice to explore interactively the virtual space.

The study gives an insight into how the author's understanding of the reconstruction process using archival drawings, but also their comprehension of the film's expressionist space from the 2D moving images, is starting to mutate due to their experience in the VR simulation. Firstly, I comprehend better the dimensions of the studio and the film sets included inside it. Solely by watching and observing the film, the dimensions of the film sets are obscure. I can estimate approximately the height and width of some of the backdrops, but the length and depth of the whole set itself are not possible to assess. However, through VR, this element can be easily defined during my virtual exploration in the studio, as I am in the studio, which is on a 1:1 scale. Even if I was aware from the

archival materials that the studio was a glasshouse, only the VR simulation allowed me to realize the amount of light entering the space, assessing how this could affect the film's shooting process. During my investigation inside the virtual reconstructed studio and setting, there was a moment when I almost saw the replicated 2D image that I had watched on the screen. This is due to my body position almost matching that of the camera's position and angle.

Having analyzed the literature and the archival materials found for *The Cabinet of Dr Caligari*, but also through the practice of recreating the film sets in VR, an analysis of the VR simulation from a first-person perspective helps to understand the benefits that this technology can offer, or cannot, to the evaluation and better understanding of the architectural space in this film. After analyzing and exploring the VR environment, I gained full awareness and comprehension of the use and construction of the backdrops for the entire film studio. The ability to explore this environment was enabled through design and reconstruction in the VR simulation.

5 LIMITATIONS AND FUTURE WORK

The purpose of the VR reconstruction discussed in this text is not to achieve a perfectly accurate representation of the film *The Cabinet of Dr. Caligari* or its sets, given the impossibility of fully replicating physical objects or understanding the atmosphere of a past film set. Nevertheless, the primary goal of the study is to simulate the closest possible representation of the film sets and studio achievable based on current investigations into the film sets and available sources. The original film setting was destroyed in the 1920s, making field-work impossible. The reconstruction relies on existing archival drawings and documents from the museums, acknowledging their potential incompleteness. Uncertainties, including incomplete date documentation and an unknown filming camera, are recognized as limitations. Despite these challenges, this research aims to provide an educated and accurate VR representation of the film's on-set environment, evaluating the effectiveness of Expressionist architecture.

This VR simulation is aimed to be oriented to both experts and the public. Therefore, the following proposed methodology could be considered by future researchers for the virtual reconstruction of other abstract, architecturally relevant film sets, such as *The Golem* (1915) by Paul Wegener and Henrik Galeen, *The Student of Prague* (1926) by Stellan Rye, *Metropolis* (1927). Nonetheless, it is essential to emphasize that, for an authentic reconstruction of any film set, as comprehensive, and complete as possible, archival drawings must be accessible. These drawings should encompass crucial data, such as scaled plans and elevations, including detailed information on dimensions and assembly instructions.

Lastly, this research proposes a methodology for preserving cultural heritage that yields significance in the history of architecture and film, presenting a first-person analysis of the VR simulation. This study explores, from the position of the researcher, the significance of VR as a research tool for understanding the film's expressionist architecture, rather than solely offering an immersive experience to users. However, future work will explore users' experience through surveys, recording their understanding of this expressionist architecture and assessing the effectiveness of this research methodology from a different angle.

6 CONCLUSIONS

By following faithfully Warm's drawings to reconstruct the virtual film sets, I discovered the similarities and differences between the reconstructed film sets and the still frames, which depict the actual setting built in 1920. Similarly, architects, scenic designers, directors, and other scholars from relevant fields can discover many elements by engaging with this VR simulation. First, they can visit an environment that existed in the 1920s, which has been destroyed, experiencing an atmosphere that they would otherwise be unable to explore. Second, they can evaluate and critically analyze the expressionist architecture by themselves. Third, they can explore how Warm's archival drawings can be used to develop the 3D reconstruction of the film sets and how accurate they can be. This type of experience has to be approached with an open mind regarding what it is possible to offer to the users as an experience itself and the need to acknowledge who the users are and what their awareness is about the film before, they enter into the VR simulation. Also, the value of this VR simulation will be different for an architect, a scenic designer, a historian, an archivist, an artist, or a film fan. The presented VR simulation depicts the reconstruction of the greatest expressionist film sets in the history of cinema which has impacted the discipline of architecture.

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